RESEARCH STATEMENT

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My research lies in gauge theory, Floer homotopy theory, and low-dimensional topology, with an emphasis on the smooth topology of four-manifolds. Dimension four exhibits distinctive phenomena, including infinitely many smooth structures on a single closed manifold and infinitely generated mapping class groups of simply connected closed manifolds. My work reveals further manifestations of this exceptional topology through two themes: developing Floer and Seiberg–Witten invariants and their homotopy refinements, and using configuration spaces and geometric techniques to study diffeomorphisms and embedded submanifolds.

Floer homotopy theory and gauge-theoretic applications. Since the fundational work of Floer [1], Kronheimer-Mrowka [2], Ozsáth-Szábo [3] and many others, various versions of Floer homologies have been powerful tools that connectes 3-manifolds with 4-manifolds. Manolescu defined the Seiberg-Witten-Floer spectra [4] for rational homology 3-spheres, which is a central tool in his disproof of the high dimensional triangulation conjecture [5]. With Khandhawit and Sasahira, we constructed the unfolded Seiberg–Witten–Floer spectra for all 3-manifolds and proved a gluing theorem for relative Bauer-Furuta invariant [6, 7]. These constructions retain equivariant information lost upon passage to homology, enabling applications of equivariant stable homotopy theory to four-manifold topology.

A remarkable application of Floer homotopy theory concerns the 11/8-conjecture: every closed smooth spin four-manifold X should satisfy $b_2(X) \geq \frac{11}{8}|\sigma(X)|$. Furuta proved $b_2(X) \geq \frac{10}{8}|\sigma(X)| + 2$ for nonzero signature by studying Pin(2)-equivariant maps between representation spheres [8]. With Hopkins, Shi, and Xu, we completely resolved the resulting Furuta–Jones problem using equivariant Mahowald invariants [9]. For $|\sigma(X)| \geq 32$, our theorem gives $b_2(X) \geq \frac{10}{8}|\sigma(X)| + c(|\sigma(X)|)$, where the explicit correction term is 4, 6, or 8. More importantly, this determines the optimal restrictions on intersection forms obtainable through this equivariant sphere-map approach.

Another application concerns stabilization. Wall's work [10] and subsequent developments show that many exotic phenomena on simply connected four-manifolds disappear after sufficiently many stabilizations (i.e., connected sums with $S^2 \times S^2$). Whether one stabilization is always enough has been an open question for many years. I proved that the exotic Dehn twist on $K3\#K3$, studied by Kronheimer and Mrowka [11], survives one stabilization: it remains topologically isotopic to the identity but not smoothly so [12]. This was the first such example on a simply connected four-manifold and its shows that one stabilization is not always enough. The key is the Pin(2)-equivariant families Bauer–Furuta invariant, whose additional equivariant information detects nontriviality even when the corresponding S^1-equivariant invariant vanishes.

I developed various refinements of families Seiberg–Witten theory to study diffeomorphism groups and moduli spaces. The gluing formula I proved confirmes McDuff's conjecture [13] that $\pi_1(\mathrm{Symp}(X)) \to \pi_1(\mathrm{Diff}(X))$ is not surjective for any closed symplectic four-manifold containing a smoothly embedded sphere of self-intersection -1 or -2 [14]. With Konno, we constructed gauge-theoretic characteristic classes for four-manifold bundles and proved that homological stability fails for the moduli spaces of every simply connected closed smooth four-manifold [15], contrasting with the low-dimensional theorem of Harer [16] and high-dimensional theorem of Galatius and Randal-Williams [17].

With Konno, Mukherjee, and Muñoz-Echániz, we connected families gauge theory with singularity theory by proving infinite-order results for boundary Dehn twists on symplectic fillings of links of minimally elliptic singularities [18]. Applications include infinite order monodromy of Milnor fibrations; compactly supported exotic diffeomorphisms of exotic $\mathbb{R}^4$'s and the first example of exotic diffeomorphisms on closed aspherical 4-manifolds [19]. More recently, families Bauer–Furuta invariants

yielded constraints on Lefschetz fibrations with four-dimensional fibers. We constructed closed simply connected symplectic four-manifolds with smoothly nontrivial Torelli symplectomorphisms, showing that their symplectic Torelli mapping class groups are not generated by squared Dehn–Seidel twists [20]. This answers Donaldson's generation question [21] negatively. (Du and Li independently obtained counterexamples using a different construction [22].)

In three papers with Ruberman and Saveliev, we studied the Seiberg–Witten invariant λ_{SW} of homology $S^1 \times S^3$'s, originally defined by Mrowka-Ruberman-Saveliev [23]. We proved a splitting formula relating it to a Lefschetz number on reduced monopole Floer homology and the Frøyshov invariant [24]. We then computed monopole Lefschetz numbers for branched-covering involutions and finite-order covering transformations [25, 26]. These formulas make λ_{SW} computable for classes of mapping tori, relate it to the Furuta–Ohta invariant [27], and yield applications to positive scalar curvature, homology cobordism, and knot concordance.

With Majewski and Rzemieniecki, we applied families gauge theory and parametrized equivariant homotopy theory to exceptional holonomy. We proved that every Cayley fibration of a compact torsion-free Spin(7)-manifold with full holonomy must have singular fibers [28]. This confirms a long-term expectation and establishes a new application of four-dimensional gauge theory to higher-dimensional geometry.

Configuration spaces, diffeomorphism groups, and submanifolds. My second direction uses configuration-space invariants to study four-dimensional isotopy and smooth structures. Watanabe disproved the four-dimensional Smale conjecture (i.e. $\mathrm{Diff}(S^4) \simeq O(5)$) using Kontsevich's configuration-space integrals [29, 30]. Motivated by a question of Freedman regarding the relation between Watanabe's work and smooth structures, Xie and I proved that these integrals depend only on a formal smooth structure: a vector bundle structure on the topological tangent microbundle [31]. The central construction is a Fulton–MacPherson compactification that requires no genuine smooth structure. Consequently, $\mathrm{Top}(4)$, the homeomorphism group of $\mathbb{R}^4$, has nonzero rational homotopy groups in infinitely many degrees, and $\mathrm{Diff}(X) \to \mathrm{Homeo}(X)$ is not rationally 2-connected for any compact oriented smooth four-manifold X. This contrasts with the homotopy equivalence of these groups in dimensions at most three.

With Wu, Xie, and Zhang, we extended the Dax invariant [32, 33] from disks to closed surfaces [34] and generalized it to the topological category [35]. These developments of the Dax theory yield many applications. In particular, for every closed oriented surface Σ of positive genus, we classified embeddings $\Sigma \hookrightarrow \Sigma \times S^2$ geometrically dual to $\{\mathrm{pt}\} \times S^2$, answering Gabai's question [36], and proved that the mapping class group of $\Sigma \times S^2$ is infinitely generated. Combining this construction with J-holomorphic curve techniques, we answered questions of McDuff–Salamon [37] and Cieliebak–Eliashberg–Mishachev [38] on symplectic 4-manifolds. On every irrational ruled surface, we found infinitely many connected components of symplectic forms in each fixed symplectic cohomology class. We also constructed infinitely many cohomologous, formally homotopic symplectic forms that are pairwise nonhomotopic through symplectic forms. These give the first such failures of connectedness and of the corresponding h-principle on closed symplectic four-manifolds.

In subsequent work with Xie and Zhang, we developed methods based on Budney–Gabai's configuration-space invariant w_3 [39]. For every compact oriented three-manifold Y with infinite fundamental group, we proved that the mapping class group of $Y \times [0,1]$ is infinitely generated [40]. We also showed that every smooth two-component split sphere link in S^4 admits infinitely many smooth splitting three-spheres that are pairwise topologically non-isotopic [41], extending Tatsuoka's theorem [42] from the unlink to arbitrarily knotted components. These results expose failures of isotopy uniqueness through configuration-space methods complementary to gauge theory.

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